

Post Graduate Professional Program (PGPP) in Data Science & Analytics

Duration: 10 Months (40 Weeks)

Mode: Online / Offline

Classes: 3 Days per Week

Class Duration: 1.5 Hours

Target Audience: Fresh Graduates, Final Year Students, Working Professionals

Program Objective

By the end of the program, students should be able to:

- Analyze real-world business datasets
- Perform data cleaning, transformation, and visualization
- Build interactive dashboards and BI reports
- Develop Machine Learning and Predictive Analytics solutions
- Work with SQL databases and data pipelines
- Build Deep Learning and Generative AI applications
- Develop RAG-based AI Assistants and AI Agents
- Deploy AI and Analytics solutions on Cloud platforms
- Work independently on enterprise level projects
- Build a professional GitHub portfolio and project portfolio

Phase 1: Programming, SQL & Data Foundation (Month 1-2)

Module 1: Python for Data Science & Analytics

Topics

- Python Fundamentals
- Variables & Data Types
- Operators
- Conditional Statements
- Loops
- Functions
- OOP Concepts
- Exception Handling
- File Handling
- Modules & Packages
- Working with APIs

Practical Work

- Student Management System
- Employee Payroll System
- Banking Management System
- Inventory Management System
- Sales Data Processing Tool

Assignment

- Build a Business Expense Tracking System

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Module 2: SQL for Data Analytics

Topics

- Database Fundamentals
- MySQL
- CRUD Operations
- Joins
- Aggregate Functions
- Group By & Having
- Views
- Stored Procedures
- Window Functions
- Query Optimization

Practical Work

- Retail Database
- Banking Database
- Hospital Database
- HR Analytics Database

Industry Case Study

- Analyze Customer Purchase Behavior using SQL

Module 3: Git & GitHub for Data Professionals

Topics

- Version Control
- Git Commands
- Branching Strategy
- Pull Requests
- Collaboration Workflow
- GitHub Portfolio Management

Practical

- Host all assignments and projects on GitHub

Phase 2: Data Analytics & Business Intelligence (Month 3-4)

Module 4: Data Analysis using Pandas & NumPy

Topics

- Data Cleaning
- Data Transformation
- Missing Value Treatment
- Feature Engineering
- Data Aggregation
- Working with APIs
- Exploratory Data Analysis

Practical Datasets

- Netflix Dataset
- IPL Dataset

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- E-Commerce Dataset
- Telecom Dataset
- Banking Dataset

Project

- Customer Churn Analysis

Module 5: Data Visualization & Business Intelligence

Tools

- Matplotlib
- Seaborn
- Plotly
- Power BI
- Tableau

Topics

- Dashboard Design
- Data Storytelling
- KPI Development
- DAX Fundamentals
- Interactive Reporting

Practical Work

- Sales Dashboard
- Customer Analytics Dashboard
- HR Analytics Dashboard
- Financial Performance Dashboard

Project

- Executive Business Intelligence Dashboard

Module 6: Statistics for Data Science

Topics

- Mean, Median, Mode
- Variance
- Standard Deviation
- Probability
- Sampling Techniques
- Hypothesis Testing
- Correlation
- Regression
- A/B Testing

Practical

- Analyze Business Datasets Statistically

Industry Case Study

- Marketing Campaign Performance Analysis

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Phase 3: Machine Learning Engineering (Month 5-6)

Module 7: Machine Learning Fundamentals

Topics

- ML Lifecycle
- Data Preprocessing
- Feature Selection
- Feature Engineering
- Train/Test Split
- Cross Validation
- Evaluation Metrics

Regression

- Linear Regression
- Multiple Regression
- Polynomial Regression

Classification

- Logistic Regression
- KNN
- Decision Trees
- Random Forest
- Naive Bayes
- SVM

Clustering

- K-Means
- Hierarchical Clustering
- DBSCAN

Projects

- House Price Prediction
- Employee Salary Prediction
- Customer Churn Prediction
- Insurance Claim Prediction

Module 8: Advanced Machine Learning

Topics

- Ensemble Learning
- Bagging & Boosting
- XGBoost
- LightGBM
- CatBoost
- Hyperparameter Tuning
- Model Explainability (SHAP)

Projects

- Credit Risk Prediction
- Loan Approval Prediction System

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- Fraud Detection Model

Phase 4: Data Engineering & Big Data (Month 7)

Module 9: Data Engineering Fundamentals

Topics

- ETL Process
- Data Warehousing
- Data Lakes
- Data Modeling
- Data Pipelines
- Apache Airflow

Practical

- Build End-to-End ETL Pipelines

Project

- Enterprise Data Warehouse Design

Module 10: Big Data Analytics

Topics

- Hadoop Ecosystem
- Apache Spark
- Spark SQL
- Distributed Computing
- Real-Time Data Processing

Projects

- Log Analytics System
- Customer Behavior Analytics
- Sales Analytics on Large Datasets

Phase 5: Deep Learning & Computer Vision (Month 8)

Module 11: Deep Learning

Topics

- Neural Networks
- Perceptrons
- Activation Functions
- Backpropagation
- TensorFlow
- Keras
- PyTorch

Practical

- Build Neural Networks from Scratch

Projects

- Customer Purchase Prediction
- Demand Forecasting Model

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Module 12: Computer Vision

Topics

- Image Processing
- OpenCV
- CNN
- Transfer Learning
- Object Detection

Projects

- Face Mask Detection
- Vehicle Detection System
- Product Defect Detection
- Crop Disease Detection

Phase 6: NLP, Generative AI & AI Agents (Month 9)

Module 13: Natural Language Processing

Topics

- Text Processing
- Tokenization
- Stemming
- Lemmatization
- POS Tagging
- Sentiment Analysis
- Named Entity Recognition

Projects

- Movie Review Sentiment Analyzer
- Spam Email Detector
- Customer Feedback Analyzer

Module 14: Generative AI & AI Agents

Topics

- LLM Fundamentals
- Prompt Engineering
- OpenAI APIs
- Hugging Face
- LangChain
- LangGraph
- AI Agents
- Function Calling

Practical

- AI Content Generator
- AI Resume Builder
- AI Email Generator
- AI Research Assistant

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Module 15: RAG & Vector Databases

Topics

- Retrieval-Augmented Generation (RAG)
- Embeddings
- ChromaDB
- Pinecone
- Vector Search
- Knowledge Bases

Projects

- PDF Chatbot
- Company Knowledge Assistant
- HR Policy Chatbot
- College Admission Assistant

Phase 7: MLOps & Cloud Deployment (Month 10)

Module 16: MLOps

Topics

- GitHub Actions
- Docker
- MLflow
- DVC
- CI/CD Basics
- Model Monitoring

Practical

- Deploy End-to-End ML Pipeline

Module 17: Cloud Deployment

Topics

- AWS Fundamentals
- EC2
- S3
- RDS
- Azure Fundamentals
- Render
- Railway
- Vercel

Practical

- Deploy AI Models Live

Project

- Production-Ready AI Application

Capstone Industry Projects (Choose One)

Banking & Finance

- Fraud Detection System

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- Credit Risk Assessment Platform
- Loan Approval Prediction AI

Retail & E-Commerce

- Product Recommendation Engine
- Customer Segmentation Platform
- Demand Forecasting System

Healthcare

- Medical Report Analyzer
- Disease Prediction System
- Patient Readmission Prediction

Human Resources

- Resume Screening AI
- Candidate Ranking System
- Employee Attrition Prediction

Education

- AI Learning Assistant
- Student Performance Predictor
- Placement Prediction System

Agriculture

- Crop Disease Detection
- Yield Prediction System
- Smart Farming Analytics

Generative AI

- Enterprise Knowledge Assistant
- Multi-Agent Research Assistant
- AI Customer Support Platform

Career Development (Throughout Program)

Technical Preparation

- Python Interview Questions
- SQL Interview Questions
- Statistics Interview Questions
- Data Analytics Interview Questions
- Machine Learning Interview Questions
- Generative AI Interview Questions

Professional Development

- Resume Building
- LinkedIn Optimization
- GitHub Portfolio Development
- Project Documentation
- Mock Interviews
- HR Interview Preparation
- Aptitude & Communication Skills

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Mandatory Portfolio Deliverables

Every student must complete:

- ✓ 25+ Assignments
- ✓ 12+ Mini Projects
- ✓ 6 Major Projects
- ✓ 1 Data Analytics Project
- ✓ 1 Business Intelligence Dashboard Project
- ✓ 1 End-to-End Machine Learning Project
- ✓ 1 Generative AI Project
- ✓ 1 RAG-Based AI Assistant
- ✓ 1 Live Industry Project
- ✓ GitHub Portfolio
- ✓ Deployed AI Application
- ✓ Professional Resume
- ✓ LinkedIn Portfolio

Industry Tools Covered

Programming & Analytics

- Python
- SQL
- Excel

Data Analytics

- NumPy
- Pandas
- SciPy

Visualization & BI

- Power BI
- Tableau
- Matplotlib
- Seaborn
- Plotly

Machine Learning

- Scikit-Learn
- XGBoost
- LightGBM
- CatBoost

Deep Learning

- TensorFlow
- Keras
- PyTorch

Data Engineering

- Apache Spark
- Airflow

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Generative AI

- OpenAI
- Hugging Face
- LangChain
- LangGraph
- ChromaDB
- Pinecone

MLOps & Cloud

- Docker
- MLflow
- DVC
- AWS
- Azure
- Render
- Railway
- Vercel

This curriculum is aligned with current industry hiring requirements for **Data Analyst, Business Analyst, Data Scientist, Machine Learning Engineer, Data Engineer, BI Developer, AI Engineer, and Generative AI Developer** roles.